SEMESTER-IV

UME518 INTRODUCTION TO ROBOTICS ENGINEERING

L	T	P	Cr
2	1	2	3.5

Course Objective: To introduce the students to the standard terminologies, applications, design specifications, and mechanical design aspects both kinematics, trajectory planning, work space analysis and work cell organisation, robot vision of industrial robotic manipulators

Syllabus

Introduction: Definition of robot, types and classifications, standard terminologies related to robotics, key design specifications used for selection of robotic manipulators for various applications, robotic applications in modern areas like automated industries, reconfigurable robots, surgical and medical robots, exoskeleton and assistive devices, flexible and soft robotics, research and non-industrial environments.

Robot Kinematics: Homogeneous coordinates and coordinate transformations; Euler angles and quaternions; Forward and Inverse kinematics for serial robotic manipulators; work space analysis, work cell organization in robotics environment, work cell design and control, Jacobian, linear and angular velocities of manipulator links.

Trajectory Planning: Robot Trajectory planning considering velocity and acceleration, joint space and Cartesian space trajectory planning, resolved motion rate control.

Robot Vision: Introduction to robot vision; Image acquisition and processing.

Mobile Robotics: Mobile robot locomotion: types, configurations and steering systems; Mapping/localization and motion planning.

Following two modules will be taught using *flip-classroom*, *self-learning* and *though guided hands-on minor projects*:

(i) **Sensors in Robotics:** Types and classification of sensors for robotic applications such as: pick and place, position/displacement, velocity, acceleration, tactile, force and torque sensors; range and proximity sensors: ultrasonic, infra-red and LASER sensors.

(ii) **Actuators and Drives in Robotics:** Linear and rotary actuators, stepper drive, DC drive, BLDC drive, Servo drive, AC Drives, hydraulic and pneumatic actuators.

(iii) **Conceptual design of robotic devices:** conceptual 3D CAD design of a *lab-scale serial robotic arm* or a *mobile robotic device* or a *robotic wrist* for a domestic/engineering or industrial application considering factors such as: Load capacity, speed of operation, positioning accuracy, work volume, self-weight, weight and rigidity of the robotic structure, types of actuators and sensors to be incorporated, power requirement and energy efficiency.

Approved in 113th meeting of the Senate held on September 7, 2024, respectively. Further, revised in 1st meeting of the Academic Council held on September 11, 2025.

Laboratory Work (if applicable)

1. Programming exercises for algorithmic implementation of forward Kinematics, inverse kinematics, motion and trajectory planning, robot vision and image-processing using programming tools like C++, MAPLE, MATLAB, ROS, Python, or MathCAD.
2. Exercises in design analysis and layout planning for the robotic workspace for generic applications

Course Learning Objectives (CLO)

The students will be able to:

1. Develop the Forward-Kinematic model/arm equation and algorithmic scheme for finding the solution for the inverse kinematics of a given serial robotic manipulator.
2. Design and analyze a robotic manipulator or develop specifications of a robotic device required for planned application/s considering its integration with other work cell devices.
3. Develop and analyze the mathematical model for a robotic controller considering trajectory planning and resolved motion rate control for a given robotic manipulator.
4. Design and implement motion planning and navigation algorithms for a mobile robotic device

Text Books

1. *Schilling, R.J., Fundamentals of Robotics Analysis and Control, Prentice Hall of India (2006).*
2. *Fu, K.S., Gonzalez, R.C. and Lee, C.S.G., Robotics: Control, Sensing, Vision, and Intelligence, McGraw Hill (1987).*
3. *Craig, J.J., Introduction to Robotics; Mechanism and control, Prentice Hall of India (2004).*
4. *Saha S.K., Introduction to Robotics, McGraw Hill, Second Edition (2014).*

Reference Books

1. *Niku S. B., Introduction to Robotics: Analysis, System, Application, Dorling Kingsley (2006).*
2. *Deb, S.R. and Deb, S., Robotics Technology and Flexible Automation, McGraw Hill (2004).*
3. *Ghoshal, A., Robotics: Fundamental Concepts and Analysis, Oxford University Press (2006).*
4. *Pratihar, D. K., Fundamental of Robotics, Alpha Science (2016).*

Evaluation Scheme:

S. No.	Evaluation Elements	Weightage (%)
1.	MST	30